

MTH 154: Percent Application: Receipts Project

Name: 

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Date: 2 December 2025

Objectives:

- (1.) I will calculate the tax and total price in a receipt given the initial price and tax rate.
- (2.) I will calculate the tax and initial price of a receipt given the tax rate and total price.
- (3.) I will calculate the initial price and total price of a receipt given the tax rate and tax.
- (4.) I will calculate the total price, the initial price, and the tax rate of a receipt given the tax, cash tendered, and remaining balance.
- (5.) I will calculate the percent of the discount and the sale price of an item given the initial price and discount.

Case 1:

Given: subtotal (\$34.10) and tax rate (1%)

Calculate: tax and total price

ASIAN FOOD	
SAMYANG JJAJANG HOT CHICKEN 5 PK [880107314236]	\$9.38
BULDAK SAMYANG XREAM	\$9.68
CARBONARA 5PK [880107314320]	\$5.98
MAYONNAISE [85444600201]	
ASIAN SNACK	
TROPICAL MANGO MARSHMALLOW [89443700008]	\$3.39
STRAWBERRY MARSHMALLOW [89443700004]	\$3.29
DRIED MANGO TAMARIND BALLS [71622105002]	\$2.38
SUBTOTAL :	\$34.10
TAX 1 (1%) :	

Tax: 1% of 34.10 is what? $.01 \times 34.10 = 0.341 \approx \0.34

Total price: $34.10 + 0.341 = 34.441 \approx \34.44

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SUBTOTAL :	\$34.10
TAX 1 (1%) :	\$0.34
TOTAL :	\$34.44

Case 2:

Given: tax rate (1%) and total (20.87)

Calculate: tax and subtotal



Let x = subtotal

$$\text{tax} = 1\% * x = 0.01x$$

$$\text{total} = x + 0.01x = 1.01x$$

$$1.01x = 20.87$$

$$x = \frac{20.87}{1.01} = 20.66336634$$

$$\text{subtotal} \cong \$20.66$$

$$\text{tax} = .01 * 20.66336634 = 0.2066336634 \cong \$0.21$$

FOOD LION
 Food Lion #0266 (540) 433-8099
 1751 South High St Harrisonburg, VA

GROCERY

SPAM LNCH MEAT 12Z	M	4.89	A *
STRKST CR TRYAKI PCH		2.18	A *
BIBIGO STICKY WHI RCE	M		
1 @ 2 FOR	6.58	3.29	A *
BIBIGO STICKY WHI RCE	M		
1 @ 2 FOR	6.58	3.29	A *

MEAT

PHILLIPS SPCL CRB MT	M	12.99	A *
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Savings

You saved:

SPAM LNCH MEAT 12Z	-1.40
PHILLIPS SPCL CRB MT	-2.00
BIBIGO STICKY WHI RCE	-2.58

Tax Paid

1.00% Tax 1	20.66	0.21
5 BALANCE DUE		20.87
Cash		50.00
CHANGE		29.13

You saved **\$5.98!**

* Year to Date

* Savings **119.07**

We value your opinion!

Tell us about your visit for a chance
 to win \$500 in free groceries.

Within the next 5 days please visit:
www.TalkToFoodLion.com

También disponible en español.
 10 winners per month
 See website for complete rules.

PIN: 0410026650070204

Loyalty Customer 46647018516

STORE: 00266 REGISTER: 007 CASHIER: 0228
 10APR2025 17:40:30
 TICKET#: 0204

www.foodlion.com
 CUSTOMER SERVICE 1-800-210-9569
 SHOPPING AT FOOD LION

Case 3:

Given: tax rate (1.00%) and tax (\$0.3866)

Calculate: subtotal and total



Let x = subtotal

$$Tax = 1\% * x = 0.01x$$

$$0.01x = 0.3866$$

$$subtotal = \frac{0.3866}{0.01} = \$38.66$$

$$total = 38.66 + 0.3866 = 39.0466 \approx \$39.05$$

FOOD LION
 Food Lion #0266 (540) 433-8099
 1751 South High St Harrisonburg, VA

FROZEN/DAIRY
 FL LG WHITE GRD A DZ W 4.99 A *
 FL WHOLE MILK 1/2GL W 2.49 A *
 FL MOZZ FNCY SHRD 2.29 A *
 TROP PP OJ W CAL 7.69 A *
 SMUCK UNCRST CHC HZL M 4.39 A *
 YPLT LF STRAWBRY 3.69 A *

MEAT
 NPR APPLWD BCN L BRD M 7.49 A *

PRODUCE
 HASS AVOCADOS W
 3 @ 1.19 3.57 A *
 LIMES W 0.33 A *
 PLUM/ROMA TOMATOES W
 0.33 lb @ 1.19/lb 0.39 A *
 YELLOW ONIONS W
 1.37 lb @ 1.19/lb 1.63 A *
 FL BABY BELLA SLICED W 2.49 A *

Savings
 You saved:
 SMUCK UNCRST CHC HZL -0.90
 NPR APPLWD BCN L BRD -1.88

Tax Paid
 1.00% Tax 1 38.66 0.39
 14 BALANCE DUE 39.05
 Cash 40.00
 CHANGE 0.95

You saved \$2.78!

*** Year to Date**
*** Savings 107.93**

 We value your opinion!
 Tell us about your visit for a chance
 to win \$500 in free groceries.
 Within the next 5 days please visit:
www.TalkToFoodLion.com
 También disponible en español.
 10 winners per month
 See website for complete rules.

PIN: 0406026630060148

 Loyalty Customer 46647018516

STORE:00266 REGISTER:006 CASHIER:0203
 TICKET#:0148 6APR2025 18:19:17
www.foodlion.com
 CUSTOMER SERVICE 1-800-210-9569
 THANK YOU FOR SHOPPING AT FOOD LION

Case 4:

Given: tax (\$3.68), cash tendered (\$40.00), and remaining balance (6.45)

Calculate: subtotal, total, and tax rate.

1 Minecraft 10pcNug L	12.99
1 10 McNuggets	
2 Nether Flame Sauce	
1 Minecraft Bundle Box	
1 L French Fries	
1 Minecraft Collectble	
1 L Diet Dr Pepper	
NO Ice	
2 Honey Must Cup	0.00
1 Minecraft Big Mac Ml	12.39
1 Big Mac	
1 Minecraft Bundle Box	
1 M French Fries	
1 Minecraft Collectble	
1 M SthSweetTea Lmnade	0.30
<Drink Upcharge>	
NO Ice	
1 Happy Meal Ch Burger	4.19
1 Cheeseburger	
1 Kids Fry	
1 Apple Slices	
1 Chocolate Milk	
1 Minecraft	
1 Spicy Buffalo Cup	0.00
Subtotal	
Tax	3.68
Take-Out Total	
Cash Tendered	40.00
Change	6.45

$$total = cash\ tendered - change$$

$$total = 40.00 - 6.45$$

$$total = 33.55$$

$$subtotal = total - tax$$

$$subtotal = 33.55 - 3.68$$

$$subtotal = 29.87$$

What % of 29.87 is 6.45?

$$\frac{\text{is}}{\text{of}} = \frac{\text{what}\%}{100}$$

$$\frac{6.45}{29.87} = \frac{\text{what}\%}{100}$$

$$\text{what}\% = \frac{6.45 * 100}{29.87}$$

$$\text{what}\% = 21.593572155\%$$

$$\text{tax \%} \approx 22\%$$

Subtotal	29.87
Tax	3.68
Take-Out Total	33.55
Cash Tendered	40.00
Change	6.45

Check

$$21.593572155\% * 29.87$$

$$0.21593572155 * 29.87 = 6.450000003 \cong \$6.45$$

Case 5:

Given: initial price and discount

Calculate: percent of the discount and sale price

FOOD LION			
Food Lion #0266 (540) 433-8099			
1751 South High St Harrisonburg, VA			
FROZEN/DAIRY			
PILLS TST DNT BST CR M			
1 @ 2 FOR 7.38		3.69 A *	
BIBIGO MINI PRK WONT M		7.99 A *	
BIBIGO MANDU CHK DMP M		7.99 A *	
GROCERY			
LARABAR CHC CHIP CKI M		10.79 A *	
LAYS CHSPKE CRB SPCE M		4.99 A *	
CouponHub Savings		-0.50 A *	
NERDS GUMMY CLTR JCY M		4.99 A *	
BARCEL TAK CRNCHY FA M		4.49 A *	
Takis Disc		-2.25 A *	
SOLELY ORG MNGO STRP M		6.89 A *	
PRODUCE			
POMEGRANATES WM			
1 @ 2 FOR 5.98		2.99 A *	
GTS SYN ORG POM LMN M		4.89 B	
CouponHub Savings		-1.00 B	
GTS SN ORG ILND BLS M		4.89 B	
Savings			
You saved:			
POMEGRANATES		-0.99	
PILLS TST DNT BST CR		-0.69	
LARABAR CHC CHIP CKI		-2.20	
LAYS CHSPKE CRB SPCE		-2.50	
NERDS GUMMY CLTR JCY		-0.50	
GTS SYN ORG POM LMN		-0.90	
GTS SN ORG ILND BLS		-0.90	
BIBIGO MINI PRK WONT		-4.00	
BIBIGO MANDU CHK DMP		-4.00	
SOLELY ORG MNGO STRP		-0.90	
Tax Paid			
1.00% Tax 1	36.28	0.36	
5.30% Tax 2	6.98	0.37	
TOTAL TAX		0.73	
11 BALANCE DUE		43.99	
Debit Card		\$43.99	

sale price = initial price – discount

discount % = 100 – what%

Larabar

sale price: $10.79 - 2.20 = \$8.59$

What percent of 10.79 is 8.59?

$$\frac{\text{is}}{\text{of}} = \frac{\text{what}\%}{100}$$

$$\frac{8.59}{10.79} = \frac{\text{what}\%}{100}$$

$$\text{what}\% = \frac{8.59 * 100}{10.79} = 79.61075075$$

$$\text{Discount}\% = 100 - 79.61075075 = 20.38924925 \approx 20\%$$

Check:

$$\text{sale} = 20.38924925\% \text{ of } 10.79$$

$$\text{sale} = 0.2038924925 * 10.79$$

$$\text{sale} = 2.199999994 \cong \$2.20$$

Lays

sale price: $4.99 - 3.00 = \$1.99$

$$\frac{\text{is}}{\text{of}} = \frac{\text{what}\%}{100}$$

$$\frac{1.99}{4.99} = \frac{\text{what}\%}{100}$$

$$\text{what}\% = \frac{1.99 * 100}{4.99} = 39.87975952$$

$$\text{Discount}\% = 100 - 39.87975952 = 60.12024048 \approx 60\%$$

Check:

$$\text{sale} = 60.12024048\% \text{ of } 4.99$$

$$\text{sale} = 0.6012024048 * 4.99$$

$$\text{sale} = \$3.00$$

Works Cited (MLA 9)

Chukwuemeka, Samuel. "Percent Applications in Receipts." *Arithmetic/Algebra Project: Percents: Show Me Your Receipts*, 2024,
conferencepresentations.appspot.com/Projects/ArithmeticAlgebra/Receipts.html.
Accessed 2 Dec. 2025.